THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 7 — MATHS · NCERT-TYPE CHAPTER TEST SET 1

Chapter 5: **Parallel and Intersecting Lines** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. If x and 24° are complementary, $x =$ 1

(a) 66° **(b)** 56° **(c)** 76° **(d)** 86°

2. In a linear pair, one angle is 2 times the other. The smaller angle is: 1

(a) 70° **(b)** 50° **(c)** 80° **(d)** 60°

3. Two parallel lines are cut by a transversal. If an interior angle is 65°, the co-interior angle on the same side is: 1

(a) 115° **(b)** 125° **(c)** 105° **(d)** 135°

4. If corresponding angles are equal, the lines are: 1

(a) perpendicular **(b)** coincident only **(c)** parallel **(d)** intersecting

5. Angle $A = 28^\circ$ and angle B are vertically opposite. Angle B is: 1

(a) 48° **(b)** 28° **(c)** 38° **(d)** 18°

6. An angle of 35° is: 1

(a) reflex **(b)** obtuse **(c)** acute **(d)** right

7. The sum of angles around a point is 360°. If three angles are 100°, 90°, 80° the fourth is: 1

(a) 90° **(b)** 100° **(c)** 110° **(d)** 80°

8. If x and 37° are complementary, $x =$ 1

(a) 43° **(b)** 53° **(c)** 73° **(d)** 63°

9. In a linear pair, one angle is 4 times the other. The smaller angle is: 1

(a) 26° **(b)** 46° **(c)** 56° **(d)** 36°

10. Two parallel lines are cut by a transversal. If an interior angle is 80°, the co-interior angle on the same side is: 1

(a) 100° **(b)** 120° **(c)** 110° **(d)** 90°

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Find x if $(2x + 10)^\circ$ and (70°) form a linear pair. 2

12. Lines $l \parallel m$ and a transversal. A pair of alternate interior angles are $(x + 10)^\circ$ and 65°. Find x. 2

13. Two angles are supplementary and one is 2 times the other. Find them. 2

14. Find x if $(2x + 20)^\circ$ and (100°) form a linear pair. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Lines $l \parallel m$, transversal t. Angle $\angle 1 = (3x + 5)^\circ$ and its corresponding angle $= (2x + 40)^\circ$. Find x and the angle. 3

16. Co-interior angles on parallel lines are $(4x + 6)^\circ$ and $(2x + 12)^\circ$. Find x and both angles. 3

17. Two lines intersect forming angles $(5x - 10)^\circ$ and $(3x + 20)^\circ$ as vertically opposite angles. Find x and all four angles. 3

OR

Lines $l \parallel m$, transversal t. Angle $\angle 1 = (3x + 10)^\circ$ and its corresponding angle $= (2x + 55)^\circ$. Find x and the angle. 3

18. Co-interior angles on parallel lines are $(4x + 10)^\circ$ and $(2x + 20)^\circ$. Find x and both angles. 3

OR

Two lines intersect forming angles $(5x - 6)^\circ$ and $(3x + 14)^\circ$ as vertically opposite angles. Find x and all four angles. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Lines $l \parallel m$ and transversal t. Angles at l: $\angle 1 = 50^\circ$. Write the values of the corresponding, alternate interior, vertically opposite and co-interior angles of $\angle 1$ at m, with reasons. 5

OR

Two parallel lines are cut by a transversal and one of the eight angles is 50°. How many of the eight angles measure 50° and how many measure 130°? Give reasons. 5

20. Lines $l \parallel m$ and transversal t. Angles at l: $\angle 1 = 65^\circ$. Write the values of the corresponding, alternate interior, vertically opposite and co-interior angles of $\angle 1$ at m, with reasons. 5

OR

Two parallel lines are cut by a transversal and one of the eight angles is 65°. How many of the eight angles measure 65° and how many measure 115°? Give reasons. 5

END OF QUESTION PAPER — Total: 40 Marks

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